

Distinct catecholaminergic pathways projecting to hippocampal CA1 transmit contrasting signals during behavior and learning

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Abstract

Neuromodulatory inputs to the hippocampus play pivotal roles in modulating synaptic plasticity, shaping neuronal activity, and influencing learning and memory. Recently it has been shown that the main sources of catecholamines to the hippocampus, ventral tegmental area (VTA) and locus coeruleus (LC), may have overlapping release of neurotransmitters and effects on the hippocampus. Therefore, to dissect the impacts of both VTA and LC circuits on hippocampal function, a thorough examination of how these pathways might differentially operate during behavior and learning is necessary. We therefore utilized 2-photon microscopy to functionally image the activity of VTA and LC axons within the CA1 region of the dorsal hippocampus in head-fixed male mice navigating linear paths within virtual reality (VR) environments. We found that within familiar environments some VTA axons and the vast majority of LC axons showed a correlation with the animals' running speed. However, as mice approached previously learned rewarded locations, a large majority of VTA axons exhibited a gradual ramping-up of activity, peaking at the reward location. In contrast, LC axons displayed a pre-movement signal predictive of the animal's transition from immobility to movement. Interestingly, a marked divergence emerged following a switch from the familiar to novel VR environments. Many LC axons showed large increases in activity that remained elevated for over a minute, while the previously observed VTA axon ramping-to-reward dynamics disappeared during the same period. In conclusion, these findings highlight distinct roles of VTA and LC catecholaminergic inputs in the dorsal CA1 hippocampal region. These inputs encode unique information, likely contributing to differential modulation of hippocampal activity during behavior and learning.

eLife assessment

This study presents **important** findings on the differential activity of noradrenergic and dopaminergic input to dorsal hippocampus CA1 in head-fixed mice traversing a runway in a virtual environment that is familiar or novel. While the data appear to be rigorously analysed, and the observed divergence in the dynamics of activity in the dopaminergic and noradrenergic axons is **solid**, there are some methodological concerns that mean the strength of evidence is currently **incomplete**.

Introduction

Catecholamines have a well established role in hippocampal function. Both dopamine and norepinephrine have been shown to impact hippocampal dependent learning and memory [Silva et al., 2012, Retaillieu and Morris, 2018, Tsetsenis et al., 2019, Hansen and Manahan-Vaughan, 2014, Gibbs and Summers, 2002, Thomas, 2015, André et al., 2015], alter synaptic plasticity [Hagena and Manahan-Vaughan, 2013, Hansen and Manahan-Vaughan, 2014, Chu et al., 2011, Edison and Harley, 2012, Hagena and Manahan-Vaughan, 2012, Goh and Manahan-Vaughan, 2013], modulate cell excitability [Edelmann and Lessmann, 2018, Segal et al., 1991], and influence the formation and stability of place cells [Kentros et al., 2004, Retaillieu and Morris, 2018], hippocampal neurons that selectively fire at specific locations in an environment [O'Keefe and Dostrovsky, 1971]. Traditionally, the main source of dopamine to the dorsal hippocampus was thought to be sparse inputs from the ventral tegmental area (VTA), while locus coeruleus (LC) inputs provided the main source of norepinephrine.

VTA DA inputs to dorsal CA1 of the hippocampus mainly innervate stratum oriens [Takeuchi et al., 2016, Adeniyi et al., 2020, Adeyelu and Ogundele, 2023], and their activity bidirectionally modulates Schaffer Collateral (CA3-CA1) synapses [Rosen et al., 2015], enhances persistence of reward-location associations [McNamara et al., 2014], and drives place preference [Mamad et al., 2017]. VTA-hippocampus input activity can also bias place field location [Mamad et al., 2017], improve place field stability across days [McNamara et al., 2014], and drive reward expectation dependent enhancement of place field quality [Krishnan et al., 2022]. However, many of the effects of DA modulation of the hippocampus have now been attributed to LC inputs as their activity enhances the strength of Schaffer Collateral synapses [Takeuchi et al., 2016], improves memory retention [Kempadoo et al., 2016], improves place field stability across days [Wagatsuma et al., 2018], and can bias place fields to a location when paired with a reward [Kaufman et al., 2020] through DA mechanisms. Although many of the effects of LC and VTA are overlapping, potentially indicating shared mechanisms of action, they are believed to play different roles in spatial learning and memory [Duszkiewicz et al., 2019]. LC inputs influence the encoding of novel environments [Kempadoo et al., 2016, Wagatsuma et al., 2018], while VTA DA inputs increase persistence of reward context associations [McNamara et al., 2014], alter hippocampal neurons firing rate [Adeniyi et al., 2020], and can bias place preference [Mamad et al., 2017]. It is possible that these differences arise because of the differences in activity observed between LC and VTA DA neurons. Therefore, characterizing the encoding properties of LC and VTA inputs directly in the hippocampus during navigation and spatial learning would provide important insights into the specific roles of these distinct neuromodulatory pathways.

Additionally, recent findings indicate considerable heterogeneity in the activity of VTA [Engelhard et al., 2019] and LC [Uematsu et al., 2017, Noei et al., 2022, Chandler et al., 2014] neurons, highlighting the need for projection specific recordings. Therefore, we functionally imaged VTA DA and LC axons with single axon resolution in dCA1 of mice as they navigated familiar and novel virtual reality (VR) environments for rewards. We observed distinct encoding properties between these sets of inputs during navigation and in response to environmental novelty. During the approach to a previously rewarded location, the activity of most VTA DA axons ramped up. In contrast, most LC axons did not show this ramping activity and instead predicted the start of motion. Additionally, a majority of LC axons and some VTA axons showed activity associated with the animal's velocity. Following exposure to a novel environment, VTA axon ramping-to-reward signals greatly reduced but LC axon activity sharply increased. These findings support distinct roles for VTA and LC inputs to the hippocampus in spatial navigation of rewarded and novel environments.

Results

To record the activity of dopaminergic inputs to the dorsal hippocampus, we expressed axonGCaMP6s or axon-GCaMP7b in LC or VTA neurons of different mice. We utilized the NET-cre mouse line [[Wagatsuma et al., 2018](#)] to restrict expression to noradrenergic LC neurons, and the DAT-Cre line [[Zhuang et al., 2005](#)] to restrict expression to dopaminergic VTA neurons (**Fig 1B**). Mice were then head-fixed and trained to run a linear virtual reality (VR) track for water rewards delivered at the end of the track (**Fig 1A**). Following reward delivery, mice were teleported to the beginning of the track and allowed to complete another lap. On experiment day mice navigated the familiar, rewarded VR environment for 10 min while 2-photon microscopy was used to image the calcium activity of LC (87 axons from 22 imaging sites in 16 mice) or VTA (7 axons from 7 imaging sites in 7 mice) axons in the dorsal CA1 (**Fig 1C**). Based on the z-axis depth of the recording planes, and the presence of increased autofluorescence in stratum pyramidal, we determined all 7 VTA axons were in *Stratum Oriens*, while for LC recordings, 18 sessions (78 axons in 11 mice) occurred in *Stratum Oriens* and 5 sessions (9 axons in 5 mice) in *Stratum Pyramidalis*. Example VTA (left, orange) and LC (right, blue) axon calcium activity aligned to the animal's behavior are shown in **Fig 1D**. Axons from both brain regions showed periodic activity linked to the animals' exploration of the VR environment.

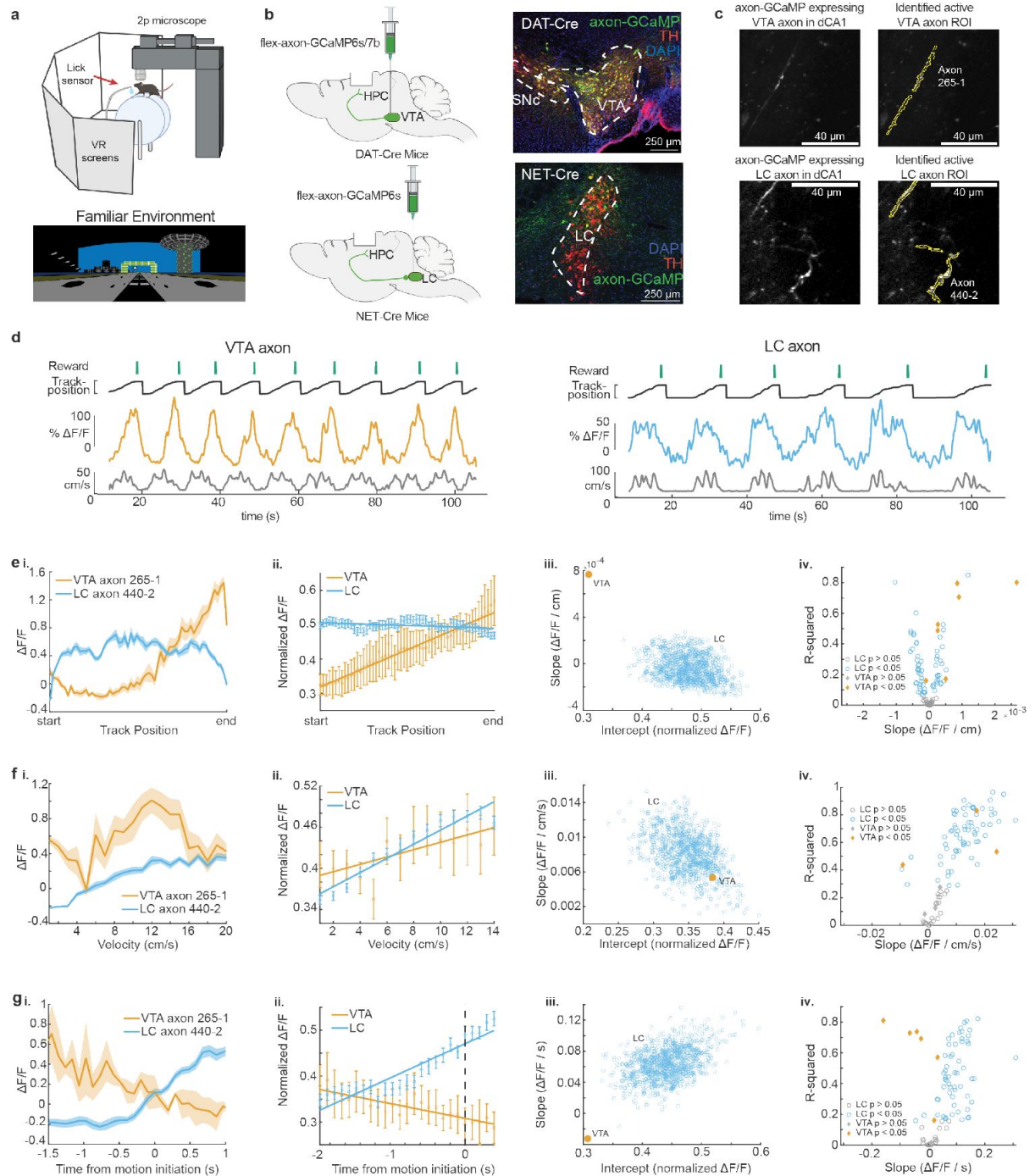


Figure 1.

Distinct activity dynamics in VTA and LC axons during navigation of familiar environments

a, Experimental setup (top), created with BioRender.com. Example virtual reality environment. **b**, Schematic representation of injection procedure (left). Representative coronal brain sections immunostained for Tyrosine Hydroxylase (TH) from a DAT-Cre mouse showing overlapping expression of axon-GCaMP (green) and TH (red) in VTA neurons (top) and from a NET-Cre mouse showing overlapping expression of axon-GCaMP (green) and TH (red) in LC neurons (bottom). **c**, Example CA1 field of view of VTA axons (top) and LC axons (bottom). Extracted regions of interest used for example VTA and LC activity throughout the figure. **d**, Example DAT-Cre mouse (left) and NET-Cre mouse (right) with aligned reward delivery (top, green), mouse track position (black), $\Delta F/F$ from example ROI (VTA-orange, LC-blue), and mouse velocity (bottom, gray). **e, i**, Position binned $\Delta F/F \pm$ s.e.m in example VTA (orange) and LC (blue) ROIs during navigation of the familiar rewarded environment. **ii**, Population average position binned $\Delta F/F \pm$ s.e.m. in VTA ROIs (orange, $n = 7$ ROIs in 7 mice) and LC ROIs (blue, $n = 87$ ROIs from 27 sessions in 17 mice). Linear regression analysis (on all data points, not means) shows that the population of VTA ROIs increase activity during approach of the end of the track while the population of LC ROIs have consistent activity throughout all positions. Linear regression, F test, VTA, $P < 1e-21$, LC, $P < 0.01$. **iii**, The LC data set was resampled 1000x using $n = 7$ axons to match the number of VTA ROIs and the slope and intercept of the regression line were measured each time (blue dots). The VTA slope is steeper than all LC slopes indicating a stronger positive relationship between position and activity for VTA inputs. **iv**, Linear regression of position binned activity of individual VTA (orange diamonds), and LC (blue, circles) axons. The majority (6/7) of VTA axons show a significant positive relationship with position while LC axons show both a positive (25/87) and negative (37/87) relationship. **f, i**, Same example ROIs as (d) binned by velocity. **ii**, Same data as (d, ii,) binned by velocity. Linear regression shows that the population of VTA and LC ROIs have a significant relationship with velocity. Linear regression, F test, VTA, $P < 0.05$, LC, $P < 1e-68$. **iii**, Resampling shows the VTA slope and intercept is within the resampled LC slopes and intercepts indicating similar relationships with velocity. **iv**, Linear regression of individual VTA and LC axons shows the majority (63/87) of LC axons have a significant positive relationship with velocity while only 2 VTA axons show this relationship. **g**, Same example ROIs as (d) aligned to motion onset. **ii**, Same data as (d, ii,) aligned to motion onset. Linear regression shows that the population of VTA axons have a negative slope prior to motion onset while LC axons have positive slope. Linear regression, F test, VTA, $P < 0.01$, LC, $P < 1e-65$. **iii**, Resampling shows the VTA slope is negative while all resampled LC slopes are positive. **iv**, Linear regression of individual VTA and LC axons shows the majority (56/87) of LC axons have a significant positive slope prior to motion onset while the majority (4/7) of VTA axons have a negative slope.

Distinct activity dynamics in VTA and LC inputs during rewarded navigation of a familiar environment

To examine axon activity further, we first looked at the mean activity across all axons as a function of track position (**Fig 1E**). As previously reported [Krishnan et al., 2022], VTA DA axons increase activity along the track, peaking at the reward location at the end of the track. In contrast, LC input activity remains relatively constant across all positions along the track (**Fig 1Eii**). To examine if this difference could be due to the lower sample size of VTA axons compared to LC axons, the LC axons were down-sampled to match the VTA sample size ($n = 7$) and the slope and intercepts of the down-sampled data was found. This was repeated 1000 times and did not generate any LC data points that overlap with VTA data demonstrating the difference in relationship between position and activity was not due to the different sample sizes (**Fig 1Eiii**). We also examined the position related activity of individual VTA and LC axons and observed a positive relationship between position and activity in 85.7% of VTA axons (6/7) but only 28.7% of LC axons (25/87) while 42.5% of LC axons (37/87) had a negative relationship between position and activity (**Fig 1E.iv**). For a subset of VTA ($n = 6$) and LC axons ($n = 26$), the reward at the end of the track was removed, and the activity of these axons in the unrewarded condition was examined

(**Fig 2A** [↗](#).) While the slope of the VTA population activity across positions significantly decreased (**Fig 2B.i** [↗](#).) as expected and previously reported [*Krishnan et al., 2022* [↗](#)], the LC population activity across positions did not significantly change (**Fig 2C.i** [↗](#).). This confirmed that the ramping activity in VTA axons was due to the animal's proximity to an expected reward and this signal was not present in the average activity of LC axons.

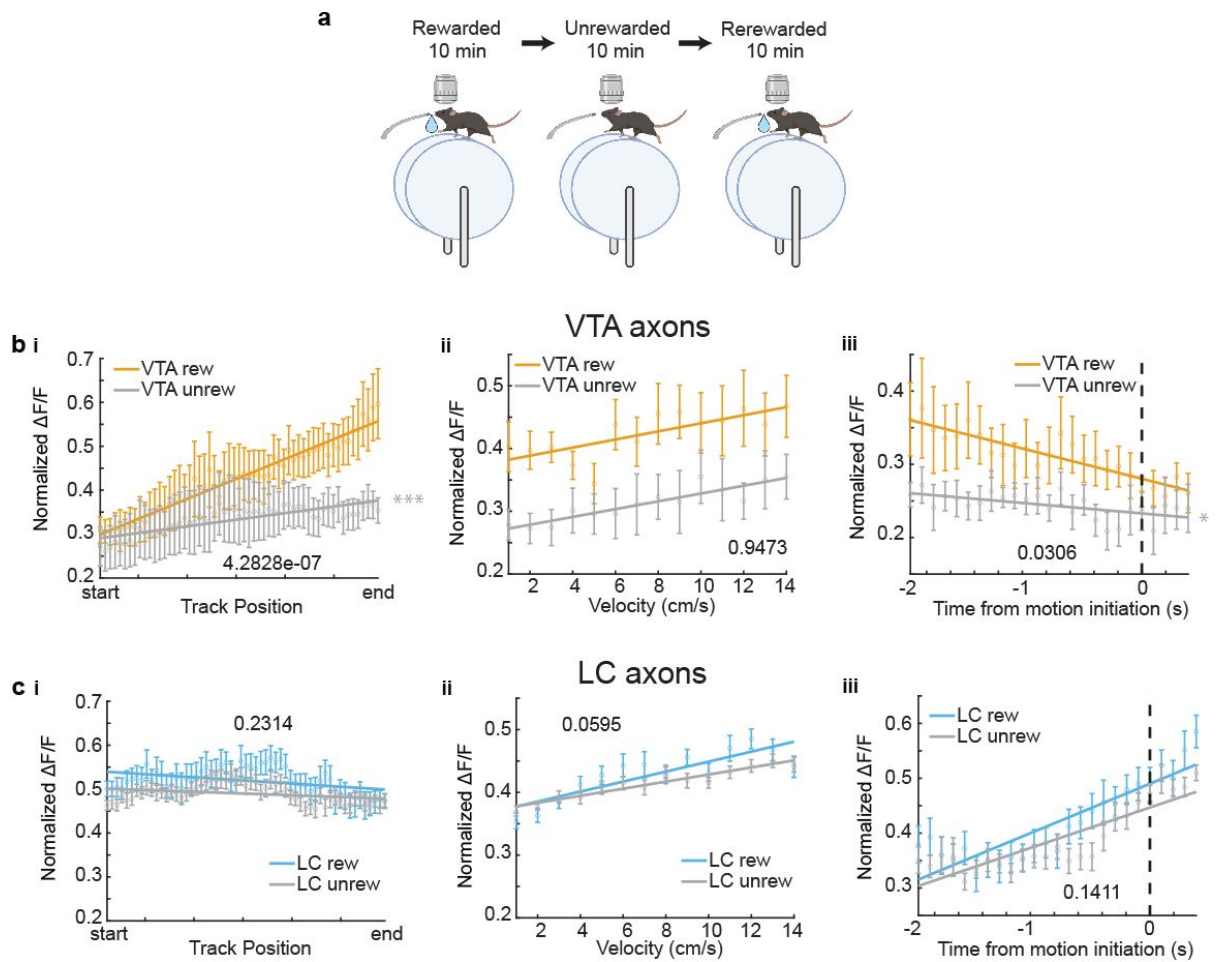


Figure 2.

Removal of reward restructures VTA but not LC input activity during spatial navigation:

a. Experimental Paradigm. **b.i.** Population average position binned $\Delta F/F \pm$ s.e.m. of VTA ROIs ($n = 6$ ROIs in 6 mice) in the rewarded (VTA rew, orange) and unrewarded conditions (VTA unrew, gray). Linear regression, F test, Rewarded, $P < 1e - 22$, Unrewarded, $P < 0.001$. **ii.** Same data as (b,i) binned by velocity. Linear regression, F test, Rewarded, $P < 0.05$, Unrewarded, $P < 0.05$. **iii.** Same data as (b,i) aligned to motion onset. Linear regression, F test, Rewarded, $P < 1e - 4$, Unrewarded, $P < 0.05$. **c, i.** Population average position binned $\Delta F/F \pm$ s.e.m. of LC ROIs ($n = 26$ ROIs in 7 sessions in 4 mice) in the rewarded (LC rew, blue) and unrewarded conditions (LC unrew, gray). Linear regression, F test, Rewarded, $P < 0.01$, Unrewarded, $P < 0.01$. **ii.** Same data as (c,i) binned by velocity. Linear regression, F test, Rewarded, $P < 1e - 12$, Unrewarded, $P < 1e - 15$. **iii.** Same data as (c,i) aligned to motion onset. Linear regression, F test, Rewarded, $P < 1e - 21$, Unrewarded, $P < 1e - 26$. The slope of each unrewarded measure was compared to the familiar laps using a one-way ANCOVA with Tukey HSD post hoc test. * $P < 0.05$, ** $P < 0.001$, *** $P < 1e - 4$.

Next, we investigated the mean activity of these axons as a function of velocity. The population mean of both VTA and LC axons increased as velocity increased (Fig 1F). This is consistent with the finding that LC inputs to dCA1 encode velocity [Kaufman et al., 2020] and the finding that DA VTA neurons encode kinematics [Engelhard et al., 2019]. Again, to account for differences in sample size, we down sampled the LC axons 1000 times and found the slope and y-intercept of each sampling. The overlap of the VTA and LC slopes and intercepts confirms we cannot conclude any differences in velocity related activity in the VTA and LC axon populations (Fig 1F.iii). However, we observed a statistically significant positive relationship between velocity and activity in the majority, 72.4%, of LC axons (63/87), while only 28.6% (2/7) of VTA axons showed a positive relationship with velocity. (Fig 1F.iv). The strong velocity correlated activity in a small subset of VTA DA axons indicates heterogeneity in the activity of these inputs similar to what is observed in VTA DA cell bodies [Engelhard et al., 2019]. To determine if the velocity correlated activity in VTA DA axons is confounded by the ramping to reward activity, we examined this activity in the unrewarded condition where the ramping to reward activity is absent. Here, the VTA population activity across all velocities is decreased but the slope, or the relationship between velocity and activity, remains unchanged (Fig 2B.ii). This is consistent with a subset of VTA axons encoding velocity information, while the overall decrease in activity is explained by the decrease in reward related activity demonstrated in Fig 2B.i. On the other hand, the relationship between velocity and LC activity is unchanged in the unrewarded environment (2C.ii), indicating condition invariant velocity encoding in LC axons.

Finally, we examined the activity of LC and VTA axons during rest and the transition to movement. The population of LC axons ramped up in activity during the 2 s leading up to motion onset (Fig 1G). This is consistent with reports of activity of LC axons in cortical areas [Reimer et al., 2016] showing LC activity prior to motion onset. In contrast, VTA axons show decreasing activity during the 2 s leading up to motion onset (Fig 1G). This ramping down in VTA axon activity is likely due to most periods of immobility occurring between reward delivery and the start of the next lap, during which we previously demonstrated reward related activity ramps down in VTA axons [Krishnan et al., 2022]. Indeed, in the unrewarded condition the negative slope of the VTA population activity leading up to motion onset disappears (Fig 2B.iii), indicating this relationship is largely driven by the presence of reward rather than motion onset. However, the LC population activity remains unchanged in the unrewarded environment (Fig 2C.iii), suggesting this activity is related to the lead up to motion initiation. These differences in activity are not an artifact of lower sample size of VTA axons as shown by down-sampling the LC axon activity 1000 times and measuring the slopes and intercepts of the down-sampled data did not generate any data points that overlapped with the VTA slope and intercept (Fig 1G.iii). In

further support of distinct activity profiles leading up to motion onset, we found that the majority (4/7) of VTA axons decreased in activity leading up to motion onset but only 2/87 LC axons decreased in activity, while the majority, (56/87), of LC axons increased in activity leading up to motion onset (**Fig 1G.iv** [↗](#)). Together, this analysis demonstrates overlapping but distinct activity in VTA and LC axons during spatial navigation with VTA axons showing strong activity correlated with distance to reward and some velocity correlated activity, while LC axons demonstrate activity correlated to velocity and time to motion onset.

Environmental Novelty induces activity in LC but not VTA inputs

Both LC and VTA neurons have been shown to respond to novel environments [*Takeuchi et al., 2016* [↗](#)]. Therefore, we aimed to examine the activity of VTA DA and LC inputs to the hippocampus during exposure to novel VR environments. Following 10 minutes in the familiar environment, mice were teleported to a novel VR environment of the same track length, with a reward at the same position at the end of the track. Following teleportation, we found the running speed of mice transiently decreases (**Fig 3A** [↗](#)) and they spend less time immobile (**Fig 3A** [↗](#)), demonstrating mice recognize they are navigating a novel environment.

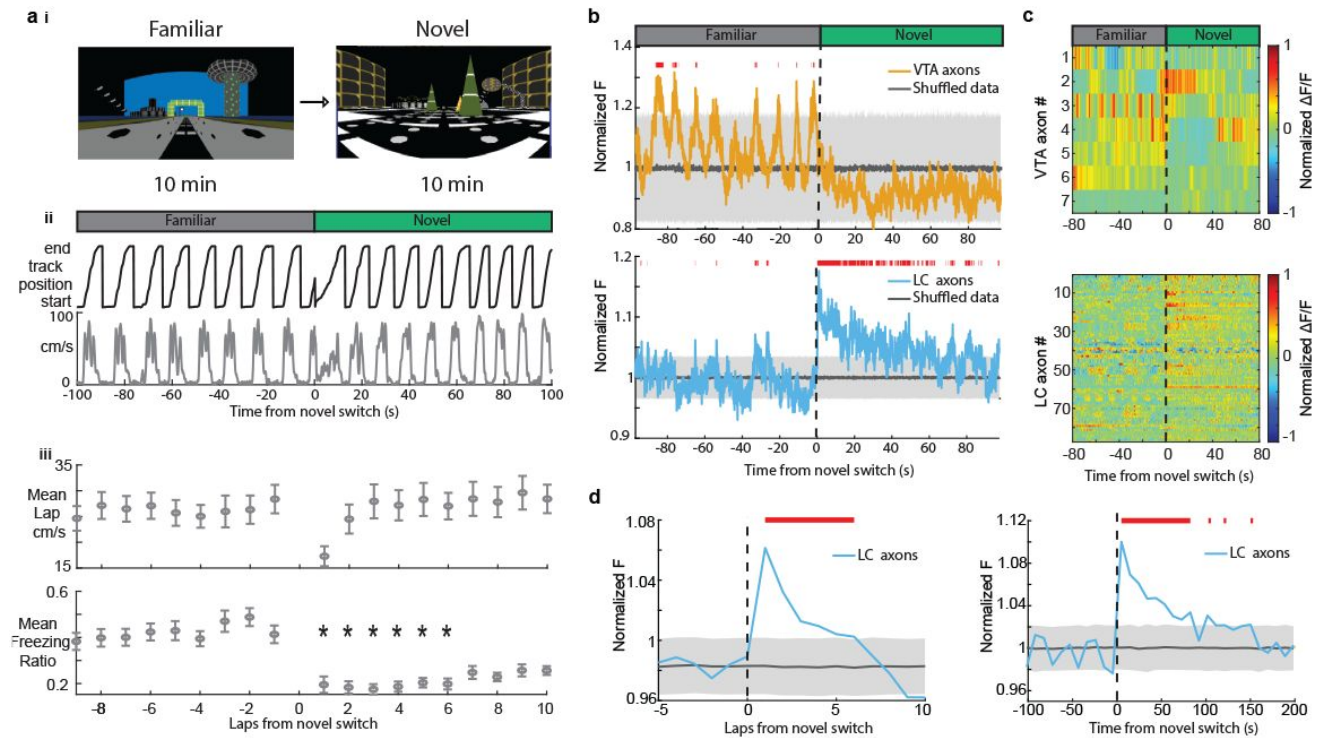


Figure 3.

Exposure to a novel environment causes an abrupt and sustained increase in activity in LC but not VTA inputs to dCA1

a, i, Experimental Paradigm. **ii**, Behavior from example mouse during the transition from the familiar VR environment to a novel VR environment showing the animals track position (top, black) and velocity (bottom, gray). **iii**, The average running velocity $\pm$ s.e.m. of all mice during the transition to a novel VR environment (top). The average freezing ratio of all mice $\pm$ s.e.m., calculated as the time spent immobile (velocity $< 5\text{cm/s}$) divided by the total lap time. Each lap was compared to the first lap in F using a one-way ANOVA with Tukey HSD post hoc test. * $P < 0.05$ **b**, Mean normalized F of all VTA ROIs (top, $n = 7$) and LC ROIs (bottom, $n = 87$) aligned to the switch to the novel environment. To define a baseline and 95% CI (gray shaded region), 1000 shuffles were created from the calcium traces and down sampled to match the sample size and averaged. This was repeated 1000 times and the mean and 95% CI of this shuffled data was determined for each frame. Red lines indicate periods where two or more consecutive frames passed above the % CI of the shuffled baseline. **c**, Normalized $\Delta F/F$ activity of all VTA ROIs (top) and LC ROIs (bottom) aligned to the switch to the novel VR environment. **d**, The normalized fluorescence of all LC ROIs binned by lap (left) or into 50 frame bins (right). The baseline and 95% CI (gray shaded region) was defined using the same method as in (b) Red lines indicate bins above the baseline 95% CI.

We aligned VTA and LC axon activity to the switch to the novel environment and investigated changes in activity due to exposure to novelty. To test whether the mean axon activity is significantly elevated or lowered, we defined a baseline by generating 1000 shuffles of the axon traces across the entire recording sessions, downsampling the shuffled data 1000 times to match the VTA ($n = 7$) and LC ($n = 87$) sample sizes, and calculating the mean and 95% CI of the shuffled data. After teleportation, the periodic activity observed in the mean of VTA axons, likely reflecting ramping-to-reward signals in each individual axon, dramatically disappears (**Fig 3B**). This is evident in the traces of most of the individual VTA axons showing a loss of the ramping-to-reward signal (**Fig 3C**), and in the VTA population position binned activity showing a significant reduction in ramping activity (**Fig 1**).

Strikingly, LC axons show a dramatic increase in mean activity that remains elevated for > 1 minute following exposure to the novel environment (**Fig 3B**) similar with findings that LC cell body activity is elevated for minutes following exposure to environmental novelty [Takeuchi et al., 2016]. A significant increase in activity above baseline activity following the switch to a novel environment can be seen in 36 LC neurons (**Fig 3C**). To further characterize this activity, we found the mean population activity for each lap and separately for 10 s time bins leading up to and following exposure to the novel environment. This analysis shows that LC activity is significantly elevated above baseline for 6 consecutive laps and approximately 90 seconds following exposure to the novel environment (**Fig 3D**). These findings demonstrate that LC inputs signal environmental novelty, supporting a role for these inputs in novelty encoding in the hippocampus.

Novelty-induced changes in behavior explains the late but not early increases in LC activity

It is possible that the change in the amount of time the animals spend running versus immobile in the novel environment could explain the increase in LC activity in the novel environment, as LC activity is related to behavior (**Fig. 1**). For instance, LC axons show elevated activity during motion versus rest (**Fig. 1G**). Therefore, an increase in the time spent in motion upon exposure to the novel environment could lead to an increase in LC activity. To account for the differences in time spent running vs immobile between the two environments, we removed any periods where the mice were immobile to isolate the effects of novelty from changes in behavior (**Fig 3A**).

When only looking at activity during running in both environments, we found that LC axon activity is elevated for 2 laps, or 40 s, in the novel environment (**Fig 4B-E** [↗](#)). This indicates that there are two separate components that drive LC axon activity during the initial exposure to the novel environment. One, a shorter purely novelty-induced increase in activity which occurs during the first 2 laps, or about 40 s, in the novel environment. Two, a behavior-induced increase in LC activity caused by an increase in the percentage of time spent running that extends beyond the increase in the novelty-induced activity for 6 laps or 90 s.

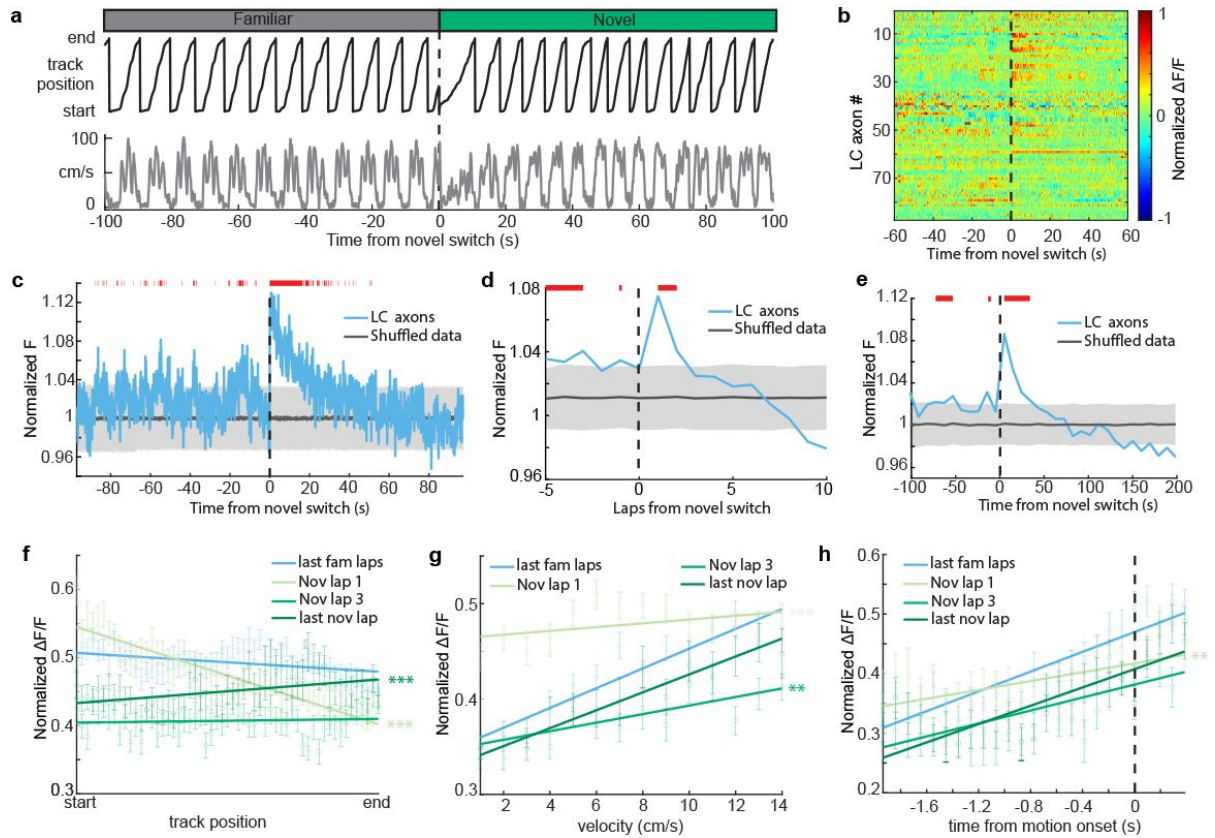


Figure 4.

Novelty-induced changes in behavior explain the late but not early increases in LC activity

a, Good behavior from example mouse following removal of freezing periods (velocity < 0.2 cm/s) during the transition from the familiar to a novel VR environment showing the animals track position (top, black) and velocity (bottom, gray). **b**, Normalized $\Delta F/F$ activity of all LC ROIs aligned to the switch to the novel VR environment following removal of freezing periods. **c**, Mean normalized F of all VTA ROIs (top, $n = 7$) and LC ROIs (bottom, $n = 90$) aligned to the switch to the novel environment. To define a baseline and 95% CI (gray shaded region), 1000 shuffles were created from the calcium traces and down sampled to match the sample size and averaged. This was repeated 1000 times and the mean and 95% CI of this shuffled data was determined for each frame. Red lines indicate periods where two or more consecutive frames passed above the 95% CI of the shuffled baseline. **d,e**, The normalized F of all LC ROIs binned by lap (d) or into 50 frame bins (e). The baseline and 95% CI (gray shaded region) was defined using the same method as in (c). Red lines indicate bins above the baseline 95% CI. **f-h**, Population average position binned (f), velocity binned (g), and motion onset aligned (h) $\Delta F/F \pm$ s.e.m. in LC ROIs ($n = 90$) in the final laps of the familiar environment (light blue), and the first (orange), third (dark blue), and final laps (green) of the novel environment. Linear regression, F test, position binned (f) fam last, $P < 1e - 4$, nov 1, $P < 1e - 68$, nov 3, $P = 0.48$, nov last, $P < 0.001$; velocity binned (g) fam last, $P < 1e - 31$, nov 1, $P < 0.05$, nov 3, $P < 1e - 5$, nov last, $P < 1e - 13$; motion onset aligned (h) fam last, $P < 1e - 29$, nov 1, $P < 0.001$, nov 3, $P < 1e - 9$, nov last, $P < 1e - 18$. The slope of each lap was compared to the final familiar laps using a one-way ANCOVA with Tukey HSD post hoc test. * $P < 0.01$, ** $P < 0.001$, *** $P < 1e - 4$.

If the short, novelty-induced signal in LC axons is an additional signal riding on top of the behavior correlated signals position, velocity, and motion onset we would expect a disruption of these behavioral correlations during the initial lap in a novel environment. To test this we examined these behavioral correlations lap-by-lap following exposure to the novel environment. Indeed, the slopes of the position binned, velocity binned, and motion onset aligned data are all significantly more negative in the first lap in the novel environment than the final laps of the familiar environment (Fig 4F-H). This is consistent with a decaying novelty signal that peaks at the start of the first lap and rides on top of these behaviorally-correlated signals. This produces an elevation in activity at positions near the start of the first lap that is lower at positions near the end of the first lap, causing a negative slope relationship between position and LC axon activity on the first lap (Fig. 4F; light green line). Further, low velocities occur at the start of each lap compared to the end of the lap. Because the novelty signal is highest when animals are running slowest, the novelty signal flattens the velocity-LC activity relationship (Fig. 4G; light green line). Lastly, rest periods typically occur at the start of the track. Therefore, motion onset encoding on the first lap in the novel environment occurs when the novelty signal is highest, again, flattening the relationship (Fig. 4H; light green line). By the third lap in the novel environment, where the novelty-induced signal is no longer observable, the relationships between LC activity and position is no longer different than the relationship in the familiar environment (Fig 4F, -green). The relationship between LC activity and motion onset is also no longer different by the third lap in the novel environment (Fig 4H). Although the relationship between velocity and LC activity is different in the third novel lap than that of the final familiar laps, by the final lap in the novel environment it is no longer different than the familiar relationship (Fig 4G). Together this demonstrates that the relationships between LC activity and behavior in the novel environment quickly return to those in the familiar environment. Interestingly, the slope is significantly increased in the final lap of the novel environment (Fig 4F), potentially indicating the development of activity at the novel reward location as has been previously reported [Kaufman et al., 2020]. Altogether, examining the lap-by-lap dynamics of the position, velocity, and motion onset activity indicates that environmental novelty induces a sharp increase in LC input activity during the first two laps in the novel environment while also inducing a change in behavior that leads to increased LC input activity in subsequent laps.

Discussion

During spatial navigation in a familiar environment, activity of VTA DA inputs to dCA1 were strongly modulated by position relative to reward, ramping up as mice approached the end of the track where reward was located. We have previously shown that this activity is dependent on the history of reward delivery and reflects the animals' reward expectation [Krishnan et al., 2022]. VTA axons in dCA1 also showed decreasing activity during rest prior to motion onset. However, after removing reward, VTA axons showed no activity prior to motion onset indicating this relationship was driven by reward related activity.

On average, VTA axons showed activity modulated by velocity in a familiar rewarded environment. This relationship was largely due to the activity of two VTA axons that were strongly modulated by velocity, suggesting that there is heterogeneity in the population of VTA axons in dCA1. A positive relationship between average VTA axon activity and velocity persisted in the unrewarded condition and in the novel environment, indicating velocity encoding in these inputs that is not a result of reward related activity. This heterogeneity in the encoding across individual VTA axons is consistent with studies demonstrating heterogeneous encoding of behavioral variables in VTA DA cell bodies, including activity related to rewards and kinematics [Engelhard et al., 2019].

Our findings that VTA DA axons show no novelty induced activity and instead show reduced activity following exposure to a novel environment, is in contrast with several studies showing novelty induced activity in VTA DA cell bodies [Takeuchi et al., 2016, Duzskiewicz et al., 2019, Lisman and Grace, 2005], indicating potential heterogeneity in VTA neurons in response to novelty. It is possible that some VTA DA inputs to dCA1 respond to novel environments, and the small number of axons recorded here are not representative of the whole population. Another possibility is that the lack of a novelty response we observe is due to differences in experimental design. Here, mice learned to approach a location for reward which has been shown to lead to ramping activity in dopaminergic VTA neurons [Howe et al., 2013, Krishnan et al., 2022, Kim et al., 2020, London et al., 2018, Jeong et al., 2022]. Following exposure to novelty, the disappearance of reward related activity could obscure any novelty induced increases in activity. Therefore, additional experiments should be conducted to investigate the heterogeneity of these axons and their activity in different experimental conditions.

LC axons showed no position encoding. Instead, they were modulated by velocity and ramped up in activity prior to motion initiation, consistent with recordings of LC axons by others in dCA1 [Kaufman et al., 2020] and in the cortex [Reimer et al., 2016], respectively. An important question is how is this LC axon activity impacting hippocampal neurons during navigation in a familiar environment? It is possible that LC axons during navigation provide increases in excitability that promotes place cell activity as both dopamine and norepinephrine in the hippocampus can impact cell excitability [Segal et al., 1991, Edelmann and Lessmann, 2011, 2018]. Additionally, place cells are flexible during spatial navigation with new place fields forming in familiar environments [Sheffield et al., 2017, Dong et al., 2021] and shifting position with time/experience [Dong et al., 2021]. Place fields can also shift to follow changing reward [Gauthier and Tank, 2018] and object locations [Bourboulou et al., 2019]. Dopamine and norepinephrine have also been shown to impact hippocampal synaptic plasticity [Zhang et al., 2009, Edelmann and Lessmann, 2011, Hagen and Manahan-Vaughan, 2012, Goh and Manahan-Vaughan, 2013]. Therefore, LC inputs may promote the plasticity necessary for place cells to flexibly adapt to changes in a familiar environment [Kaufman et al., 2020, Redondo and Morris, 2011]. In other words, LC inputs could allow the hippocampus to be flexible during navigation through their impacts on synaptic plasticity [Takeuchi et al., 2016, Yamasaki and Takeuchi, 2017, Duzskiewicz et al., 2019].

Exposure to environmental novelty leads to an increase in dopamine in the dorsal hippocampus [*Ihalainen et al., 1999*] and promotes synaptic plasticity [*Li et al., 2003*, *Hagena and Manahan-Vaughan, 2012*], hippocampal replay [*McNamara et al., 2014*, *Dupret et al., 2010*] and memory persistence [*Li et al., 2003*, *Cohen et al., 2017*]. In our experiment, exposure to a novel environment caused an increase in LC axon activity but not in VTA DA axon activity, supporting findings that novel experiences induce activity of LC neurons [*Takeuchi et al., 2016*]. The activity of LC neurons in turn increases hippocampal neuron activity [*Wagatsuma et al., 2018*], increases efficacy of Schaffer Colateral synapses [*Takeuchi et al., 2016*], stabilizes place cells across days [*Wagatsuma et al., 2018*] and memory persistence [*Wagatsuma et al., 2018*, *Takeuchi et al., 2016*, *Chowdhury et al., 2022*] through dopamine receptor dependent mechanisms. Importantly, while LC inputs to CA1 have been shown to cause an increase in activity [*Wagatsuma et al., 2018*, *Chowdhury et al., 2022*], shape over-representation of novel reward locations [*Kaufman et al., 2020*], and modulate memory linking [*Chowdhury et al., 2022*], they have not been shown to play a role in the formation of contextual memories [*Chowdhury et al., 2022*] or stabilization of place cell maps across days [*Wagatsuma et al., 2018*] in dCA1. However, it is possible this novelty induced activity in LC inputs to dCA1 impacts the formation of instant place fields observed in novel environments [*Sheffield et al., 2017*, *Dong et al., 2021*]. Instant place fields form on the first lap of a novel environment, right when the LC novelty signal is highest in dCA1, suggesting LC inputs may play a role in their formation or stabilization. Therefore, further experiments should investigate the role of LC axons on dCA1 place fields on a trial-by-trial basis.

While LC neurons have been shown to impact novelty encoding through dopaminergic mechanisms [*Wagatsuma et al., 2018*, *Takeuchi et al., 2016*, *Chowdhury et al., 2022*], this does not exclude the possibility that they also release norepinephrine during exposure to novelty and exploration of a familiar environment. Indeed, hippocampal levels of norepinephrine also increase during exposure to environmental novelty [*Lima et al., 2019*, *Moreno-Castilla et al., 2017*], but how this norepinephrine release effects hippocampal function is not well understood. Additionally, it is not known whether norepinephrine and dopamine are released from the same LC inputs or from distinct sets of LC inputs. Dopamine is in the synthesis pathway of norepinephrine and in LC neurons it is loaded into vesicles where it is then converted to norepinephrine by dopamine β -hydroxylase [*Cimarusi et al., 1979*]. It is possible that high levels of activity of LC inputs, like those occurring during exposure to novelty, lead to release of vesicles before dopamine can be converted to norepinephrine thus leading to the release of dopamine under these conditions. However, low levels of LC activation, like those observed during familiar environment exploration, may provide time for dopamine to be converted to norepinephrine and thus lead to the release of norepinephrine from LC terminals under these conditions. Further experiments investigating the dynamics of dopamine conversion and release from LC terminals in the hippocampus should be conducted to test this hypothesis.

Here we show that LC input activity is modulated by velocity, time to motion onset, and exposure to novel environments. Each of these conditions is associated with an increase in arousal, and LC activity has been strongly linked to arousal levels [*Aston-Jones and Bloom, 1981*, *Berridge and Waterhouse, 2003*, *Carter et al., 2010*]. Therefore, rather than encoding each of these variables independently, LC inputs are likely encoding the animals arousal level during spatial navigation. It has been shown that attention and arousal levels impact tuning properties in many cortical areas and this is thought to be mediated through LC activity [*Shulman et al., 1979*, *Bouret and Sara, 2002*, *Martins and Froemke, 2015*, *Waterhouse and Navarra, 2019*]. Similarly, changes in the animals' brain state, including changes in attention [*Kentros et al., 2004*] and engagement [*Pettit et al., 2022*], alter the tuning properties of place cells. This indicates arousal could impact the function of hippocampal neurons through these LC inputs, either directly or through astrocytes [*Peter Rupprecht et al., 2023*].

The distinct activity dynamics exhibited by LC and VTA DA axons during spatial navigation of familiar and novel environments underscore their distinct contributions to hippocampal dependent learning and memory processes. Notably, these findings reinforce the notion that VTA DA inputs play a pivotal role in the ongoing maintenance and updating of associations between expected rewards and the locations that lead to them, while LC axons appear to be integral to the process of encoding memories of entirely new environments and stimuli.

Methods

Subjects

All experimental and surgical procedures were in accordance with the University of Chicago Animal Care and Use Committee guidelines. For this study, we used 10–20 week-old male *Slc6a3Cre⁺/(DAT-Cre⁺/-)* mice and *Slc6a2 Cre⁺/(NET-Cre⁺/-)* (23–33 g). Male mice were used over female mice due to the size and weight of the headplates (9.1 mm × 31.7 mm, 2 g) which were difficult to firmly attach on smaller female skulls. Mice were individually housed in a reverse 12 h light/dark cycle at 72 °F and 47% humidity, and behavioral experiments were conducted during the animal's dark cycle.

Mouse surgery and viral injections

Mice were anesthetized (1–2% isoflurane) and injected with 0.5 ml of saline (intraperitoneal injection) and 0.05 ml of Meloxicam (1–2 mg/kg, subcutaneous injection) before being weighed and mounted onto a stereotaxic surgical station (David Kopf Instruments). A small craniotomy (1–1.5 mm diameter) was made over the ventral tegmental area (VTA) ($\pm$ 0.5 mm lateral, 3.1mm caudal of Bregma) of *DAT-Cre⁺/-* mice or over the locus coeruleus (LC) ($\pm$ 0.875 mm lateral, -5.45 mm caudal of Bregma). The genetically-encoded calcium indicator, pAAV-hsyn-Flex-Axon-GCaMP6s (pAAV-hSynapsin1-FLEX-axon-GCaMP6s was a gift from Lin Tian (Addgene viral prep # 112010-AAV5 ; <http://n2t.net/addgene:112010> ; RRID:Addgene 112010) was injected into the VTA of *DAT-Cre⁺/-* mice (200 nL at a depth of 4.4 mm below the surface of the dura) or the LC of *NET-Cre⁺/-* mice (200 nL at a depth of 3.65 mm below Bregma). For a subset (4/7) of VTA recordings, a different GCaMP variant, pAAV-Ef1A-Flex-Axon-GCaMP7b, was injected due to the difficulty finding and recording VTA axons in dCA1 (pAAV-Ef1a-Flex-Axon-GCaMP7b (pAAV-Ef1a-Flex-Axon-GCaMP7b was a gift from Rylan Larsen -Addgene plasmid # 135419; <http://n2t.net/addgene:135419> ; RRID: Addgene 135419). Following injections, the site was covered up using dental cement (Metabond, Parkell Corporation) and a metal head-plate (9.1 mm × 31.7 mm, Atlas Tool and Die Works) was also attached to the skull with the cement. Mice were separated into individual cages and water restriction began 3 weeks later (0.8–1.0 ml per day). Mice then underwent surgery to implant a hippocampal window as previously described [Dombeck et al., 2010]. Following implantation, the headplate was attached with the addition of a head-ring cemented on top of the head-plate which was used to house the microscope objective and block out ambient light. Post-surgery mice were given 2–3 ml of water/day for 3 days to enhance recovery before returning to the reduced water schedule (0.8–1.0 ml/day).

Behavior and virtual reality

Our virtual reality (VR) and treadmill setup was designed similar to previously described setups [Sheffield et al., 2017, Heys et al., 2014]. The virtual environments that the mice navigated through were created using VIRMEn [Aronov and Tank, 2014]. 2 m (*DAT-Cre* mice) or 3 m (*NET-Cre* mice) linear tracks rich in visual cues were created that evoked numerous place fields in mice as they moved along the track at all locations (Fig. 1) [Bourboulou et al., 2019]. Mice were head restrained with their limbs comfortably resting on a freely rotating styrofoam wheel ('treadmill'). Movement of the wheel caused movement in VR by using a rotatory encoder to detect treadmill rotations and feed this information into our VR computer, as in refs. [Sheffield et al.,

[2017](#), [Heys et al., 2014](#)]. Mice received a water reward (4 μ L) through a waterspout upon completing each traversal of the track (a lap), which was associated with a clicking sound from the solenoid. Licking was monitored by a capacitive sensor attached to the waterspout. Upon receiving the water reward, a short VR pause of 1.5 s was implemented to allow for water consumption and to help distinguish laps from one another rather than them being continuous. Mice were then virtually teleported back to the beginning of the track and could begin a new traversal. Mouse behavior (running velocity, track position, reward delivery, and licking) was collected using a PicoScope Oscilloscope (PICO4824, Pico Technology, v6.13.2). Behavioral training to navigate the virtual environment began 4–7 days after window implantation (30 min per day) and continued until mice reached >2 laps per minute, which took 10–14 days (although some mice never reached this threshold). Mice that reached this behavioral threshold were imaged the following day.

Two-photon imaging

Imaging was done using a laser scanning two-photon microscope (Neurolabware). Using a 8 kHz resonant scanner, images were collected at a frame rate of 30 Hz with bidirectional scanning through a 16x/0.8 NA/3 mm WD water immersion objective (MRP07220, Nikon). GCaMP6s and GCaMP7b were excited at 920 nm with a femtosecond pulsed two photon laser (Insight DS + Dual, Spectra-Physics) and emitted fluorescence was collected using a GaAsP PMT (H11706, Hamamatsu). The average power of the laser measured at the objective ranged between 50–80 mW. A single imaging field of view (FOV) between 400–700 μ m equally in the x/y direction was positioned to collect data from as many VTA or LC axons as possible. Time-series images were collected from 3–5 planes spaced 2 μ m apart using an electric lens to ensure axons remained in a field of view and reduce power going to an individual plane. Images were collected using Scanbox (v4.1, Neurolabware) and the PicoScope Oscilloscope (PICO4824, Pico Technology, v6.13.2) was used to synchronize frame acquisition timing with behavior.

Imaging sessions

The familiar environment was the same environment that the animals trained in. The experiment protocol for single day imaging sessions is shown in [Fig. 1](#). Each trial lasted 8–12 min and was always presented in the same order. Mice were exposed to the familiar rewarded environment for 10 minutes, then were immediately teleported to the start of a novel rewarded VR environment and allowed to navigate for 10 minutes. Mice on average ran 19 ± 3.8 (mean $\pm$ 95% CI) laps in the familiar environment, at which point they were teleported to the novel environment and imaging continued for 30 ± 5.4 laps. The Novel-rewarded environment (N) had distinct visual cues, colors and visual textures, but the same dimensions (2 or 3 m linear track) and reward location (end of the track) as the familiar environment. Imaging sessions with large amounts of drift or bleaching were excluded from analysis (8 sessions for NET mice, 6 sessions for LC Mice).

Histology and brain slices imaging

We checked the expression axon-GCaMP to confirm expression was restricted to the VTA of DAT-Cre mice and the LC of NET-Cre. Mice were deeply anesthetized with isoflurane and perfused with 10 ml phosphate-buffered saline (PBS) followed by 20 mL 4% paraformaldehyde in PBS. The brains were removed and immersed in 30% sucrose solution overnight before being sectioned at 30 μ m thickness on a cryostat. Brain slices were collected into well plates containing PBS. Slices were washed 5 times with PBS for 5 min then were blocked in 1% Bovine Serum Albumin, 10% Normal goat serum, 0.1% Triton X-100 for 2 h. Brain slices were then incubated with 1:500 rabbit- α -TH (MAB318, Sigma Aldrich) and 1:500 mouse- α -GFP (SAB2702197, Sigma Aldrich) in blocking solution at 4 °C. After 48 h, the slices were incubated with 1:1000 goat- α -rabbit Alexa Fluor 647nm secondary antibody (A32731, ThermoFisher) and 1:1000 goat- α -mouse Alexa Fluor 488nm (A32723, ThermoFisher) for 2 h. Brain slices were then collected on glass slides and mounted with a mounting media with DAPI (SouthernBiotech DAPI-Fluoromount-G Clear Mounting Media,

010020). The whole-brain slices were imaged under $\times 10$ and $\times 40$ with a Caliber I.D. RS-G4 Large Format Laser Scanning Confocal microscope from the Integrated Light Microscopy Core at the University of Chicago.

Image processing and ROI selection

Time-series images were preprocessed using Suite2p (v0.10.1)79. Movement artifacts were removed using rigid and non-rigid transformations and assessed to ensure absence of drifts in the z-direction. Datasets with visible z-drifts were discarded (8 sessions). For axon imaging, ROIs were first defined using Suite2p and manually inspected for accuracy. ROIs were then hand drawn over all segments of Suite2p defined active axons using ImageJ to ensure all axon segments were included for analysis. Fluorescent activity for each ROI was extracted and highly correlated ROIs (Pearson correlation coefficient ≥ 0.7) were combined and their fluorescent activity was extracted. To be included Baseline corrected $\Delta F/F$ traces across time were then generated for each ROI using both a small window of 300 frames for lap by lap analysis, and a larger sliding window of 2000 frames to avoid flattening slow signals for novelty response analysis. Additional ROIs were drawn over autofluorescent structures that were not identified by suite2p. These “blebs” were processed in the same way as axon ROIs and used as controls to check for imaging and motion artifacts.

To remove low signal to noise axons, we defined the SNR of each ROI using the power spectrum of their fluorescent activity similar to [Reimer et al., 2016]. For frequencies above 1Hz, the power was defined as noise because this sits outside of the range of frequencies possible for GCaMP6s fluorescence. The SNR ratio was then defined as the ratio of the peak power between 0.5 Hz and 1Hz over the average power between 1 Hz and 3 Hz. The SNR of “blebs” was also determined and any axon with a SNR greater than 1.5 std from the mean of the “blebs” SNR was used for analysis (110/231 LC ROIs, 7/7 VTA ROIs).

Additionally, it was observed that a subset of axon ROIs would greatly increase fluorescence at seemingly random timepoints and remain elevated for the rest of the trial. This activity could be due to the axons being unhealthy and filling with calcium. Therefore, we identified these axons using the *cusum* function in matlab to detect changes in mean activity that remained elevated for at least 2000 frames or at least 500 frames if they were still elevated at the end of the recording session and removed them from analysis (20/110 LC ROIs, 0/7 VTA ROIs).

Behavioral Analysis

Mouse velocity was calculated as the change in VR position divided by the sampling rate and smoothed using a Savitzky-Golay filter with a 7 frame window and 5 degree polynomial. To find the lap mean velocity, periods where the mice were immobile (velocity < 5 cm/s) were removed and the average velocity during the remaining frames was calculated. The lap mean freezing ratio was calculated as the number of frames spent immobile (velocity < 0.2 cm/s) divided by the total number of frames for each lap.

Axon Imaging Analysis

For the three measures below, to avoid weighting axons with a high SNR more than others each ROI was normalized by $(\Delta F/F - \Delta F/F \min)/(\Delta F/F \max - \Delta F/F \min)$ where $\Delta F/F \min$ is the 1st quantile and $\Delta F/F \max$ was the 99 quantile for each ROI. The 1st and 99th quantiles were used in order to avoid normalize to noisy outlier data points.

Position binned fluorescence

To find the position binned fluorescent activity of each ROI, the track was divided into 5 cm bins. For each lap, the average fluorescence in each bin was calculated for each ROI. The position binned fluorescence was then averaged across all laps in each environment to find the mean position binned activity in the familiar and novel environments.

Velocity binned fluorescent activity

To find the velocity binned fluorescent activity for each ROI, the velocity was divided into 1 cm/s bins from 1 to 30 cm/s. Velocities above this 14 cm/s were excluded from figures because not all mice ran faster than 14 cm/s. For each lap, the ROIs average fluorescence in each velocity bin was calculated and then averaged across all laps in each environment to find the velocity binned activity in the familiar and novel environments.

Motion initiation aligned fluorescence

Periods where mice were immobile (velocity < 5 cm/s) for at least 1.5s then proceed to run (velocity ≥ 5cm/s) for at least 3s were identified. The fluorescent activity for ROIs for these periods was aligned to the frame mice began running (velocity crossed above 5cm/s). The average aligned fluorescent activity of each ROI was then determined for each environment.

Linear regression analysis

To assess dynamics between each of the above measures and calcium activity of LC and VTA axons, we performed linear regression on the population's familiar environment data and significance was assessed with an F test. To compare the dynamics between LC and VTA axons, we performed exact testing based on Monte-Carlo resampling (1000 resamples with sample size matching the lower sample size condition) as detailed in legends ([Fig 1E](#) ).

To assess the changing position and velocity encoding of LC axons following exposure to a novel environment, we performed linear regression on the population fluorescence data of the average of the last 4 laps in the familiar environment, and each of the first three laps in the novel environment for each measure. The significance for the fit of each line was assessed with an F test, and an ANCOVA was conducted to test for differences in slope between the four laps. The same process was conducted for the motion initiation dynamics, but only using ROIs in mice who paused within the first 2 laps and 30s following exposure to the novel environment.

Novel response analysis

To examine the response of LC and VTA axons to the novel VR environment, the fluorescence data was normalized by the mean for each ROIs and aligned to the frame where the mice were switched to the novel environment and the mean normalized F for LC and VTA ROIs at each time point was calculated. Baseline fluorescent activity was then calculated for LC and VTA ROIs separately by generating 1000 shuffled traces of the ROIs calcium activity and subsampling down to the sample size (90 for LC; 7 for VTA) 1000 times and finding the mean of the subsampled shuffles. The mean and 95% CI of all 1000 subsamples was found and the mean activity of LC and VTA ROIs was considered significantly elevated when it passed above the 95% CI of the shuffled data. The same process was repeated to define a baseline for the time binned data (fluorescent activity divided into 50 frame bins) and the lap binned data (mean activity for each lap).

Additionally, to account for changes in behavior between the familiar and novel environments, periods where the animals were immobile (velocity ≤ 0.2 cm/s) were removed and running periods were concatenated together and aligned to the switch to the novel environment. Here, we again defined a baseline for the time mean traces, time binned activity, and the lap binned activity using the above bootstrapping approach.

Figure graphics

All figure graphics including [Fig 1A-B](#)  were created using BioRender.com.

Code availability

Scripts used for data analysis are available on Github https://github.com/chadheer/LC_VTA_paper 

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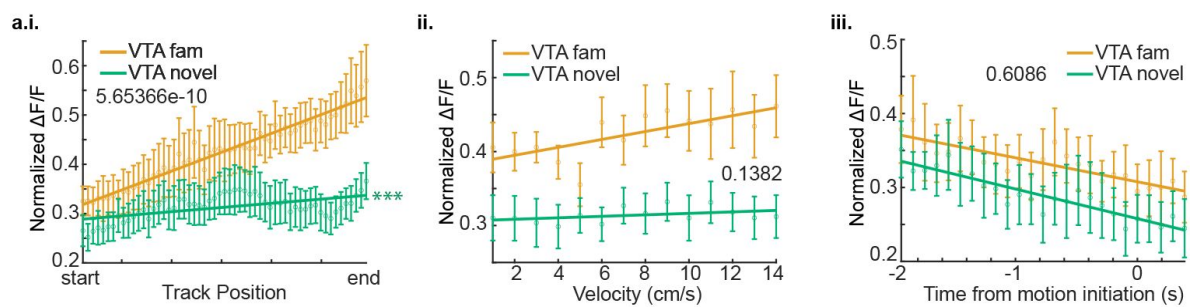
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Author contributions

C.H. and M.S. conceived and designed the experiments. C.H. collected and analyzed the data. C.H. and M.S. interpreted the data and wrote the manuscript.

Declaration of interests

The authors declare no competing interests.



Supplementary Figure 1.

a,i, Population average position binned $\Delta F/F \pm$ s.e.m. of VTA ROIs ($n = 7$ ROIs in 7 mice) in the familiar (orange) and novel (green) rewarded environments. Linear regression, F test, Rewarded, $P < 1e - 21$, Unrewarded, $P < 0.01$. **ii,** Same data as (a,i) binned by velocity. Linear regression, F test, Rewarded, $P < 0.05$, Unrewarded, $P = 0.57$. **iii,** Same data as (a,i) aligned to motion onset. Linear regression, F test, Rewarded, $P < 0.01$, Unrewarded, $P < 0.001$. The slope of each novel measure was compared to the familiar laps using a one-way ANCOVA with Tukey HSD post hoc test. * $P < 0.05$, ** $P < 0.001$, *** $P < 1e - 4$.

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Reviewer #1 (Public Review):

Summary:

Heer and Sheffield used 2 photon imaging to dissect the functional contributions of convergent dopamine and noradrenaline inputs to the dorsal hippocampus CA1 in head-restrained mice running down a virtual linear path. Mice were trained to collect water rewards at the end of the track and on test days, calcium activity was recorded from dopamine (DA) axons originating in the ventral tegmental area (VTA, $n=7$) and noradrenaline axons from the locus coeruleus (LC, $n=87$) under several conditions. When mice ran laps in a familiar environment, VTA DA axons exhibited ramping activity along the track that correlated with distance to reward and velocity to some extent, while LC input activity remained constant across the track, but correlated invariantly with velocity and time to motion onset. A subset of recordings taken when the reward was removed showed diminished ramping activity in VTA DA axons, but no changes in the LC axons, confirming that DA axon activity is locked to reward availability. When mice were subsequently introduced to a new environment, the ramping to reward activity in the DA axons disappeared, while LC axons showed a dramatic increase in activity lasting 90 s (6 laps) following the environment switch. In the final analysis, the authors sought to disentangle LC axon activity induced by novelty vs. behavioral changes induced by novelty by removing periods in which animals were immobile and established that the activity observed in the first 2 laps reflected novelty-induced signal in LC axons.

Strengths:

The results presented in this manuscript provide insights into the specific contributions of catecholaminergic input to the dorsal hippocampus CA1 during spatial navigation in a rewarded virtual environment, offering a detailed analysis of the resolution of single axons. The data analysis is thorough and possible confounding variables and data interpretation are carefully considered.

Weaknesses:

Aspects of the methodology, data analysis, and interpretation diminish the overall significance of the findings, as detailed below.

The LC axonal recordings are well-powered, but the DA axonal recordings are severely underpowered, with recordings taken from a mere 7 axons (compared to 87 LC axons). Additionally, 2 different calcium indicators with differential kinetics and sensitivity to calcium changes (GCaMP6S and GCaMP7b) were used ($n=3$, $n=4$ respectively) and the data pooled. This makes it very challenging to draw any valid conclusions from the data, particularly in the novelty experiment. The surprising lack of novelty-induced DA axon activity may be a false negative. Indeed, at least 1 axon (axon 2) appears to be showing a novelty-induced rise in activity in Figure 3C. Changes in activity in 4/7 axons are also referred to as a 'majority' occurrence in the manuscript, which again is not an accurate representation of the observed data.

The authors conducted analysis on recording data exclusively from periods of running in the novelty experiment to isolate the effects of novelty from novelty-induced changes in behavior. However, if the goal is to distinguish between changes in locus coeruleus (LC) axon activity induced by novelty and those induced by motion, analyzing LC axon activity during periods of immobility would enhance the robustness of the results.

The authors attribute the ramping activity of the DA axons to the encoding of the animals' position relative to reward. However, given the extensive data implicating the dorsal CA1 in timing, and the remarkable periodicity of the behavior, the fact that DA axons could be signalling temporal information should be considered.

The authors should explain and justify the use of a longer linear track (3m, as opposed to 2m in the DAT-cre mice) in the LC axon recording experiments.

<https://doi.org/10.7554/eLife.95213.1.sa2>

Reviewer #2 (Public Review):

Summary:

The authors used 2-photon Ca^{2+} -imaging to study the activity of ventral tegmental area (VTA) and locus coeruleus (LC) axons in the CA1 region of the dorsal hippocampus in head-fixed male mice moving on linear paths in virtual reality (VR) environments.

The main findings were as follows:

- In a familiar environment, the activity of both VTA axons and LC axons increased with the mice's running speed on the Styrofoam wheel, with which they could move along a linear track through a VR environment.
- VTA, but not LC, axons showed marked reward position-related activity, showing a ramping-up of activity when mice approached a learned reward position.
- In contrast, the activity of LC axons ramped up before the initiation of movement on the Styrofoam wheel.
- In addition, exposure to a novel VR environment increased LC axon activity, but not VTA axon activity.

Overall, the study shows that the activity of catecholaminergic axons from VTA and LC to dorsal hippocampal CA1 can partly reflect distinct environmental, behavioral, and cognitive factors. Whereas both VTA and LC activity reflected running speed, VTA, but not LC axon activity reflected the approach of a learned reward, and LC, but not VTA, axon activity reflected initiation of running and novelty of the VR environment.

I have no specific expertise with respect to 2-photon imaging, so cannot evaluate the validity of the specific methods used to collect and analyse 2-photon calcium imaging data of axonal activity.

Strengths:

- (1) Using a state-of-the-art approach to record separately the activity of VTA and LC axons with high temporal resolution in awake mice moving through virtual environments, the authors provide convincing evidence that the activity of VTA and LC axons projecting to dorsal CA1 reflect partly distinct environmental, behavioral and cognitive factors.
- (2) The study will help a) to interpret previous findings on how hippocampal dopamine and norepinephrine or selective manipulations of hippocampal LC or VTA inputs modulate behavior and b) to generate specific hypotheses on the impact of selective manipulations of hippocampal LC or VTA inputs on behavior.

Weaknesses:

- (1) The findings are correlational and do not allow strong conclusions on how VTA or LC inputs to dorsal CA1 affect cognition and behavior. However, as indicated above under

Strengths, the findings will aid the interpretation of previous findings and help to generate new hypotheses as to how VTA or LC inputs to dorsal CA1 affect distinct cognitive and behavioral functions.

(2) Some aspects of the methodology would benefit from clarification.

First, to help others to better scrutinize, evaluate, and potentially to reproduce the research, the authors may wish to check if their reporting follows the ARRIVE (Animal Research: Reporting of In Vivo Experiments) guidelines for the full and transparent reporting of research involving animals (<https://arriveguidelines.org/>). For example, I think it would be important to include a sample size justification (e.g., based on previous studies, considerations of statistical power, practical considerations, or a combination of these factors). The authors should also include the provenance of the mice. Moreover, although I am not an expert in 2-photon imaging, I think it would be useful to provide a clearer description of exclusion criteria for imaging data.

Second, why were different linear tracks used for studies of VTA and LC axon activity (from line 362)? Could this potentially contribute to the partly distinct activity correlates that were found for VTA and LC axons?

Third, the authors seem to have used two different criteria for defining immobility.

Immobility was defined as moving at <5 cm/s for the behavioral analysis in Figure 3a, but as <0.2 cm/s for the imaging data analysis in Figure 4 (see legends to these figures and also see Methods, from line 447, line 469, line 498)? I do not understand why, and it would be good if the authors explained this.

(3) In the Results section (from line 182) the authors convincingly addressed the possibility that less time spent immobile in the novel environment may have contributed to the novelty-induced increase of LC axon activity in dorsal CA1 (Figure 4). In addition, initially (for the first 2-4 laps), the mice also ran more slowly in the novel environment (Figure 3aIII, top panel). Given that LC and VTA axon activity were both increasing with velocity (Figure 1F), reduced velocity in the novel environment may have reduced LC and VTA axon activity, but this possibility was not addressed. Reduced LC axon activity in the novel environment could have blunted the novelty-induced increase. More importantly, any potential novelty-induced increase in VTA axon activity could have been masked by decreases in VTA axon activity due to reduced velocity. The latter may help to explain the discrepancy between the present study and previous findings that VTA neuron firing was increased by novelty (see Discussion, from line 243). It may be useful for the authors to address these possibilities based on their data in the Results section, or to consider them in their Discussion.

(4) Sensory properties of the water reward, which the mice may be able to detect, could account for reward-related activity of VTA axons (instead of an expectation of reward). Do the authors have evidence that this is not the case? Occasional probe trials, intermixed with rewarded trials, could be used to test for this possibility.

<https://doi.org/10.7554/eLife.95213.1.sa1>

Reviewer #3 (Public Review):

Summary:

Heer and Sheffield provide a well-written manuscript that clearly articulates the theoretical motivation to investigate specific catecholaminergic projections to dorsal CA1 of the hippocampus during a reward-based behavior. Using 2-photon calcium imaging in two groups of cre transgenic mice, the authors examine the activity of VTA-CA1 dopamine and LC-CA1 noradrenergic axons during reward seeking in a linear track virtual reality (VR) task. The authors provide a descriptive account of VTA and LC activities during walking, approach to reward, and environment change. Their results demonstrate LC-CA1 axons are activated

by walking onset, modulated by walking velocity, and heighten their activity during environment change. In contrast, VTA-CA1 axons were most activated during the approach to reward locations. Together the authors provide a functional dissociation between these catecholamine projections to CA1. A major strength of their approach is the methodological rigor of 2-photon recording, data processing, and analysis approaches. These important systems neuroscience studies provide solid evidence that will contribute to the broader field of learning and memory. The conclusions of this manuscript are mostly well supported by the data, but some additional analysis and/or experiments may be required to fully support the author's conclusions.

Weaknesses:

(1) During teleportation between familiar to novel environments the authors report a decrease in the freezing ratio when combining the mice in the two experimental groups (Figure 3a-iii). A major conclusion from the manuscript is the difference in VTA and LC activity following environment change, given VTA and LC activity were recorded in separate groups of mice, did the authors observe a similar significant reduction in freezing ratio when analyzing the behavior in LC and VTA groups separately?

(2) The authors satisfactorily apply control analyses to account for the unequal axon numbers recorded in the LC and VTA groups (e.g. Figure 1). However, given the heterogeneity of responses observed in Figures 3c, 4b and the relatively low number of VTA axons recorded (compared to LC), there are some possible limitations to the author's conclusions. A conclusion that LC-CA1 axons, as a general principle, heighten their activity during novel environment presentation, would require this activity profile to be observed in some of the axons recorded in most all LC-CA1 mice. Additionally, if the general conclusion is that VTA-CA1 axons ramp activity during the approach to reward, it would be expected that this activity profile was recorded in the axons of most all VTA-CA1 mice. Can the authors include an analysis to demonstrate that each LC-CA1 mouse contained axons that were activated during novel environments and that each VTA-CA1 mouse contained axons that ramped during the approach to reward?

(3) A primary claim is that LC axons projecting to CA1 become activated during novel VR environment presentation. However, the experimental design did not control for the presentation of a familiar environment. As I understand, the presentation order of environments was always familiar, then novel. For this reason, it is unknown whether LC axons are responding to novel environments or environmental change. Did the authors re-present the familiar environment after the novel environment while recording LC-CA1 activity?

<https://doi.org/10.7554/eLife.95213.1.sa0>

Author Response

Reviewer #1 (Public Review):

Summary:

Heer and Sheffield used 2 photon imaging to dissect the functional contributions of convergent dopamine and noradrenaline inputs to the dorsal hippocampus CA1 in head-restrained mice running down a virtual linear path. Mice were trained to collect water rewards at the end of the track and on test days, calcium activity was recorded from dopamine (DA) axons originating in the ventral tegmental area (VTA, n=7) and noradrenaline axons from the locus coeruleus (LC, n=87) under several conditions. When mice ran laps in a familiar environment, VTA DA axons exhibited ramping activity along

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The reviewer points out a weakness in the analysis of VTA axons in our dataset. The relatively low n (currently 7) comes from the fact that VTA axons in the CA1 region of the hippocampus are very sparse and very difficult to record from (due to their sparsity and the low level of baseline fluorescence inherent in long range axon segments). This is the reason they have not been recorded from in any other lab outside of our lab. LC axons, on the other hand, are more abundant in CA1. In the paper when comparing VTA versus LC axons we deal with the mismatch in n by downsampling the LC axons to match the VTA axons and repeated this 1000 times to create a distribution. However, because the VTA axon n is relatively low, it is possible that we have not sampled the VTA axon population sufficiently and therefore have a biased population in our dataset. The issue is that it takes months for the baseline expression of GCaMP to reach sufficient levels to be able to record from VTA axons, and it is typical to find only a single axon in a FOV per animal. There are additional reasons why mice and/or axon recordings do not reach criteria and cannot be included in the dataset (these exclusion criteria are reported in the Methods section). For instance, out of the 54 DAT-Cre mice injected, images were never conducted in 36 for lack of expression or because mice failed to reach behavioral criteria. Another 11 mice were excluded for heat bubbles that developed during imaging, z-drift of the FOV, or bleaching of the GCaMP signal.

However, we do have $n=2$ additional VTA axon recordings that we will add to the dataset to bring the n up from 7 to 9. We plan on re-analyzing the data with $n=9$ VTA axons and making

comparisons to down-sampled LC axons as described above. This boost in n will increase the power of our VTA axon analysis. To more formally test whether this is sufficient for statistical tests, we plan to utilize the G*power power-analysis tool to compute statistical power for each of the different tests we use. We will report this in the next version of the paper. However, the $n=2$ additional axons were not recorded in the novel environment, so the next version will remain at $n=7$ for the novel environment analysis. We agree with the reviewer that the lack of the novelty induced DA axon activity may be a false negative, and so we will adjust the description of our results and discussion accordingly.

During the data collection of VTA axon activity we tried two variants of GCaMP: 6s and 7b, to see if one would increase the success rate of finding and recording from VTA axons. Given the long time-course of these experiments and the low yield in success, we pooled the GCaMP variants together to increase statistical power. Because the 2 additional VTA DA axons that were recorded from expressed GCaMP6s, the next version of the paper will have $n=5$ GCaMP6s, and $n=4$ GCaMP7b VTA DA axons, which will allow us to compare the activity of the two sensors in the familiar environment. The reviewer correctly pointed out that the sensors themselves could confound our results, and so they should not be pooled unless we can show they do not produce different signals in the axons. We will make this comparison and report the findings in the next version of the paper. If we find no significant differences, we will pool the data. If differences are detected, we will keep these axons separate for subsequent analysis and comparisons to LC axons.

The authors conducted analysis on recording data exclusively from periods of running in the novelty experiment to isolate the effects of novelty from novelty-induced changes in behavior. However, if the goal is to distinguish between changes in locus coeruleus (LC) axon activity induced by novelty and those induced by motion, analyzing LC axon activity during periods of immobility would enhance the robustness of the results.

This is indeed true, and this suggested analysis could further support our conclusions regarding the LC novelty signal. For the next version of the paper, we will use the periods of immobility to analyze and isolate any novelty induced activity in LC axons. However, following exposure to the novel environment, mice spend much less time immobile, therefore there may not be sufficient periods of immobility close in time to the exposure to the novel environment (which is when the novelty signal occurs). We plan to analyze mouse behavior during the early exposure to the novel environment for immobility and check whether we have enough of this behavior to perform the suggested analysis.

The authors attribute the ramping activity of the DA axons to the encoding of the animals' position relative to reward. However, given the extensive data implicating the dorsal CA1 in timing, and the remarkable periodicity of the behavior, the fact that DA axons could be signalling temporal information should be considered.

This is a very good point. We agree that the VTA DA axons could be signaling temporal information, as we have previously shown that these axons also exhibit ramping activity when you average their activity by time to reward (Krishnan et al., 2022). We will conduct this analysis on this dataset. We have not, however, conducted any experiments designed to separate out time from distance, such as the experiments conducted in Kim et al., 2020. Therefore, we cannot determine whether this is due to proximity in space to reward or time to reward. We will clarify in our text that by proximity, we mean either place or time, and cannot conclude which feature of the experience drives the VTA axon signal.

Krishnan, L.S., Heer, C., Cherian, C., Sheffield, M.E. Reward expectation extinction restructures and degrades CA1 spatial maps through loss of a dopaminergic reward proximity signal. Nat Commun 13, 6662 (2022).

Kim, HyungGoo R., Athar N. Malik, John G. Mikhael, Pol Bech, Iku Tsutsui-Kimura, Fangmiao Sun, Yajun Zhang, et al. A Unified Framework for Dopamine Signals across Timescales. *Cell* 183, no. 6 (2020).

The authors should explain and justify the use of a longer linear track (3m, as opposed to 2m in the DAT-cre mice) in the LC axon recording experiments.

LC axon activity was recorded on a 3m track to match the track length from an experiment we recently published (Dong et al., 2021) in which mice were exposed to a novel 3m track while populations of CA1 pyramidal cells were recorded. In that paper we described the time course of place field formation on the novel track. We wanted to test if LC axons signaled novelty (as we hypothesized) and whether the time course of LC axon activity matched the time course of place field formation. We briefly discuss this in the Discussion section of this paper and hypothesize that LC axons in CA1 could open a window of plasticity in which new place fields can form.

VTA axons were recorded on a 2m track (same VR tracks as LC axons were recorded on) to match another recent paper from our lab in which reward expectation was manipulated (Krishnan et al, 2022). In that study CA1 populations of pyramidal cells were recorded during the reward expectation experiment. To match the experience during recordings of VTA axons in CA1 to test how reward expectation may influence axon signaling along the track, we also used a 2m track. The idea was to check how VTA dopaminergic inputs to CA1 may influence CA1 population dynamics along the track.

Although the tracks were identical for LC and VTA recordings for both the familiar and novel tracks in terms of visual cues and design, the track lengths are different (simply modulated by gain control of the rotary encoder). To account for this we normalized the lengths for our comparison analysis. This normalization allows for a direct comparison of the patterns of activity across the two types of axons, controlling for the potential confound introduced by the different track lengths. By adjusting the data to a common scale, we could assess the relative changes in activity levels at matched spatial bins, ensuring that any observed differences or similarities are due to the intrinsic properties of the axons rather than differences in track lengths. However, the different lengths do make the animal's experience slightly different. This is somewhat offset by the observations in our study that none of the LC or VTA axon signals would be expected to be majorly influenced by variations in track length. For instance, LC axons are associated with velocity and a pre-motion initiation signal, neither of which would be influenced by track length. VTA axons are also associated with velocity, which would not influence a direct comparison to LC axon velocity signals as mice reach maximal velocity very rapidly along the track. VTA axons do ramp up in activity as they approach the reward zone, and this signal could be modulated by track length (or maybe not if the signal is encoding time to reward rather than distance). However, LC axons show no ramping to reward signals, so a comparison across axons recorded on different track lengths for this analysis is justified.

However, to add rigor to comparisons of axon dynamics recorded along 2m and 3m tracks, we plan to plot axon activity of both sets of axons by time to reward, and actual (un-normalized) distance from reward.

Krishnan, L.S., Heer, C., Cherian, C., Sheffield, M.E. Reward expectation extinction restructures and degrades CA1 spatial maps through loss of a dopaminergic reward proximity signal. *Nat Commun* 13, 6662 (2022).

Dong, C., Madar, A. D. & Sheffield, M.E. Distinct place cell dynamics in CA1 and CA3 encode experience in new environments. *Nat Commun* 12, 2977 (2021).

Reviewer #2 (Public Review):

Summary:

The authors used 2-photon Ca^{2+} -imaging to study the activity of ventral tegmental area (VTA) and locus coeruleus (LC) axons in the CA1 region of the dorsal hippocampus in head-fixed male mice moving on linear paths in virtual reality (VR) environments.

The main findings were as follows:

- *In a familiar environment, the activity of both VTA axons and LC axons increased with the mice's running speed on the Styrofoam wheel, with which they could move along a linear track through a VR environment.*
- *VTA, but not LC, axons showed marked reward position-related activity, showing a ramping-up of activity when mice approached a learned reward position.*
- *In contrast, the activity of LC axons ramped up before the initiation of movement on the Styrofoam wheel.*
- *In addition, exposure to a novel VR environment increased LC axon activity, but not VTA axon activity.*

Overall, the study shows that the activity of catecholaminergic axons from VTA and LC to dorsal hippocampal CA1 can partly reflect distinct environmental, behavioral, and cognitive factors. Whereas both VTA and LC activity reflected running speed, VTA, but not LC axon activity reflected the approach of a learned reward, and LC, but not VTA, axon activity reflected initiation of running and novelty of the VR environment.

I have no specific expertise with respect to 2-photon imaging, so cannot evaluate the validity of the specific methods used to collect and analyse 2-photon calcium imaging data of axonal activity.

Strengths:

(1) Using a state-of-the-art approach to record separately the activity of VTA and LC axons with high temporal resolution in awake mice moving through virtual environments, the authors provide convincing evidence that the activity of VTA and LC axons projecting to dorsal CA1 reflect partly distinct environmental, behavioral and cognitive factors.

(2) The study will help a) to interpret previous findings on how hippocampal dopamine and norepinephrine or selective manipulations of hippocampal LC or VTA inputs modulate behavior and b) to generate specific hypotheses on the impact of selective manipulations of hippocampal LC or VTA inputs on behavior.

Weaknesses:

(1) The findings are correlational and do not allow strong conclusions on how VTA or LC inputs to dorsal CA1 affect cognition and behavior. However, as indicated above under Strengths, the findings will aid the interpretation of previous findings and help to generate new hypotheses as to how VTA or LC inputs to dorsal CA1 affect distinct cognitive and behavioral functions.

(2) Some aspects of the methodology would benefit from clarification. First, to help others to better scrutinize, evaluate, and potentially to reproduce the research, the authors may wish to check if their reporting follows the ARRIVE (Animal Research: Reporting of In Vivo Experiments) guidelines for the full and transparent reporting of research involving animals (<https://arriveguidelines.org/>). For example, I

think it would be important to include a sample size justification (e.g., based on previous studies, considerations of statistical power, practical considerations, or a combination of these factors). The authors should also include the provenance of the mice. Moreover, although I am not an expert in 2-photon imaging, I think it would be useful to provide a clearer description of exclusion criteria for imaging data.

We thank the reviewer for helping us formalize the scientific rigor of our study. There are ten ARRIVE Guidelines and we have addressed most of them in our study already. However, there is an opportunity to add detail. We have listed below all ten points and how we have or will address each one.

(1) Experimental design - we go into great depth explaining the experimental set-up, how we used the autofluorescent blebs as imaging controls, how we controlled for different sample sizes between the two populations, and the statistical tests used for comparisons. We also carefully accounted for animal behavior when quantifying and describing axon dynamics both in the familiar and novel environments.

(2) Sample size - We state both the number of ROIs and mice for each analysis. Wherever we state how many axons had a certain kind of activity, we will also state the number of mice we saw this activity in. For the next version of the paper, we plan to conduct a power analysis using G*power to assess the power of our sample sizes for statistical analysis.

(3) Inclusion/exclusion criteria - Out of the 36 NET-Cre mice injected, 15 were never recorded for either failing to reach behavioral criteria, or a lack of visible expression in axons. Out of the 54 DAT-Cre mice injected, images were never conducted in 36 for lack of expression or failing to reach behavioral criteria. Out of the remaining 21 NET-CRE, 5 were excluded for heat bubbles, z-drift, or bleaching, while 11 DAT-Cre were excluded for the same reasons. This was determined by visually assessing imaging sessions, followed by using the registration metrics output by suite2p. This registration metric conducted a PCA on the motion-corrected ROIs and plotted the first PC. If the PC drifted largely, to the point where no activity was apparent, the video was excluded from analysis.

(4) Randomization - Already included in the paper is a description of random down sampling of LC axons to make statistical comparisons with VTA axons. LC axons were selected pseudo-randomly (only one axon per imaging session) to match VTA sampling statistics. This randomization was repeated 1000 times and comparisons were made against this random distribution.

(5) Blinding-masking - no blinding/masking was conducted as no treatments were given that would require this. We will include this statement in the next version.

(6) Outcomes - We defined all outcomes measured, such as those related to animal behavior and related axon signaling.

(7) Statistical methods - None of the reviewers had any issues regarding our description of statistical methods, which we described in detail in this version of the paper.

(8) Experimental animals - We described that DAT- Cre mice were obtained through JAX labs, and NET-Cre mice were obtained from the Tonegawa lab (Wagatsuma et al. 2017)

(9) Experimental procedure - Already listed in detail in Methods section.

(10) Results - Rigorously described in detail for behaviors and related axon dynamics.

Wagatsuma, Akiko, Teruhiro Okuyama, Chen Sun, Lillian M. Smith, Kuniya Abe, and Susumu Tonegawa. "Locus Coeruleus Input to Hippocampal CA3 Drives Single-Trial Learning of a

Novel Context.” Proceedings of the National Academy of Sciences 115, no. 2 (January 9, 2018): E310–16. <https://doi.org/10.1073/pnas.1714082115>.

Second, why were different linear tracks used for studies of VTA and LC axon activity (from line 362)? Could this potentially contribute to the partly distinct activity correlates that were found for VTA and LC axons?

A detailed response to this is written above for a similar comment from reviewer 1.

Third, the authors seem to have used two different criteria for defining immobility. Immobility was defined as moving at <5 cm/s for the behavioral analysis in Figure 3a, but as <0.2 cm/s for the imaging data analysis in Figure 4 (see legends to these figures and also see Methods, from line 447, line 469, line 498)? I do not understand why, and it would be good if the authors explained this.

This is an error leftover from before we converted velocity from rotational units of the treadmill to cm/s. This will be corrected in the next version of the paper.

(3) In the Results section (from line 182) the authors convincingly addressed the possibility that less time spent immobile in the novel environment may have contributed to the novelty-induced increase of LC axon activity in dorsal CA1 (Figure 4). In addition, initially (for the first 2-4 laps), the mice also ran more slowly in the novel environment (Figure 3aIII, top panel). Given that LC and VTA axon activity were both increasing with velocity (Figure 1F), reduced velocity in the novel environment may have reduced LC and VTA axon activity, but this possibility was not addressed. Reduced LC axon activity in the novel environment could have blunted the novelty-induced increase. More importantly, any potential novelty-induced increase in VTA axon activity could have been masked by decreases in VTA axon activity due to reduced velocity. The latter may help to explain the discrepancy between the present study and previous findings that VTA neuron firing was increased by novelty (see Discussion, from line 243). It may be useful for the authors to address these possibilities based on their data in the Results section, or to consider them in their Discussion.

This is a great point. The decreased velocity in the novel environment could lead to a diminished novelty response in LC axons. We will add a discussion point on this in the next version. This could also be the case for VTA axons, so will add a discussion point that the lack of novelty signaling seen in VTA axons could be due to reduced velocity masking this signal.

(4) Sensory properties of the water reward, which the mice may be able to detect, could account for reward-related activity of VTA axons (instead of an expectation of reward). Do the authors have evidence that this is not the case? Occasional probe trials, intermixed with rewarded trials, could be used to test for this possibility.

Mice receive their water reward through a waterspout that is immobile and positioned directly in front of their mouth (which is also immobile as they are head fixed) and water delivery is triggered by a solenoid when the mice reach the end of the virtual track. Therefore, because the waterspout remains in the same place relative to the mouse, and the water reward is not delivered until they reach the end of the virtual track, there is nothing for the mice to detect. We will update the paper to make this clearer.

Additionally, on the initial laps with no reward, the ramping activity is still present (Krishnan et al, 2022) indicating this activity is not directly related to the presence/absence of water but is instead caused by reward expectation.

Reviewer #3 (Public Review):

Summary:

Heer and Sheffield provide a well-written manuscript that clearly articulates the theoretical motivation to investigate specific catecholaminergic projections to dorsal CA1 of the hippocampus during a reward-based behavior. Using 2-photon calcium imaging in two groups of cre transgenic mice, the authors examine the activity of VTA-CA1 dopamine and LC-CA1 noradrenergic axons during reward seeking in a linear track virtual reality (VR) task. The authors provide a descriptive account of VTA and LC activities during walking, approach to reward, and environment change. Their results demonstrate LC-CA1 axons are activated by walking onset, modulated by walking velocity, and heighten their activity during environment change. In contrast, VTA-CA1 axons were most activated during the approach to reward locations. Together the authors provide a functional dissociation between these catecholamine projections to CA1. A major strength of their approach is the methodological rigor of 2-photon recording, data processing, and analysis approaches. These important systems neuroscience studies provide solid evidence that will contribute to the broader field of learning and memory. The conclusions of this manuscript are mostly well supported by the data, but some additional analysis and/or experiments may be required to fully support the author's conclusions.

Weaknesses:

(1) During teleportation between familiar to novel environments the authors report a decrease in the freezing ratio when combining the mice in the two experimental groups (Figure 3a-iii). A major conclusion from the manuscript is the difference in VTA and LC activity following environment change, given VTA and LC activity were recorded in separate groups of mice, did the authors observe a similar significant reduction in freezing ratio when analyzing the behavior in LC and VTA groups separately?

In response to this comment, we will analyze the freezing ratios in DAT-Cre and NET-Cre mice separately. However, other members of the lab have seen the same result in other mouse strains (See Dong et al. 2021), so we do not expect to see a difference (but it is certainly worth checking).

(2) The authors satisfactorily apply control analyses to account for the unequal axon numbers recorded in the LC and VTA groups (e.g. Figure 1). However, given the heterogeneity of responses observed in Figures 3c, 4b and the relatively low number of VTA axons recorded (compared to LC), there are some possible limitations to the author's conclusions. A conclusion that LC-CA1 axons, as a general principle, heighten their activity during novel environment presentation, would require this activity profile to be observed in some of the axons recorded in most all LC-CA1 mice.

We agree with the reviewer's point here. To help avoid this problem, when downsampling LC axons to compare to VTA axons, we matched the sampling statistics of the VTA axons/mice (i.e. only one LC axon was taken from each mouse to match the VTA dataset).

However, in the next version of the paper we will also report the number of mice that we see a significant novel response in. We will also add the number of mice with significant activity for each of the measures in the familiar environment (e.g. how many mice had axons positively correlated with velocity).

Additionally, if the general conclusion is that VTA-CA1 axons ramp activity during the approach to reward, it would be expected that this activity profile was recorded in the

axons of most all VTA-CA1 mice. Can the authors include an analysis to demonstrate that each LC-CA1 mouse contained axons that were activated during novel environments and that each VTA-CA1 mouse contained axons that ramped during the approach to reward?

As stated above, we will add the number of mice that had each activity type we reported here.

(3) A primary claim is that LC axons projecting to CA1 become activated during novel VR environment presentation. However, the experimental design did not control for the presentation of a familiar environment. As I understand, the presentation order of environments was always familiar, then novel. For this reason, it is unknown whether LC axons are responding to novel environments or environmental change. Did the authors re-present the familiar environment after the novel environment while recording LC-CA1 activity?

This is an important point to address. While we never varied the presentation order of the familiar vs novel environments, we did record the activity of LC axons in some of the mice in a dark environment (no VR cues) prior to exposure to the familiar environment. We will look at these axons to address whether they respond to initial exposure to the familiar environment. This will allow us to check whether they are responding to environmental change or novelty. We will add this analysis to the next version of the paper.